



CYBER SECURITY TASK - 2

SOC Alert Monitoring & Incident Response Simulation Report

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Executive Summary

This report provides a detailed Security Operations Center (SOC) alert monitoring and incident response analysis based on simulated logs ingested into Splunk Enterprise. The purpose is to identify suspicious events, classify incidents, and outline necessary remediation steps following standard SOC procedures.

Tools

1. Splunk Enterprise 10.0.2 Free Trial
2. Sample alert logs

Environment Setup

SIEM Tool: Splunk Enterprise 10.x (Free Trial)

- Data Source: SOC_Standard_Logs (100 log events)
- Log Types: Authentication logs, Network connection logs, Malware detection events
- Objective: Detect anomalies, classify alerts, and document an incident response workflow.



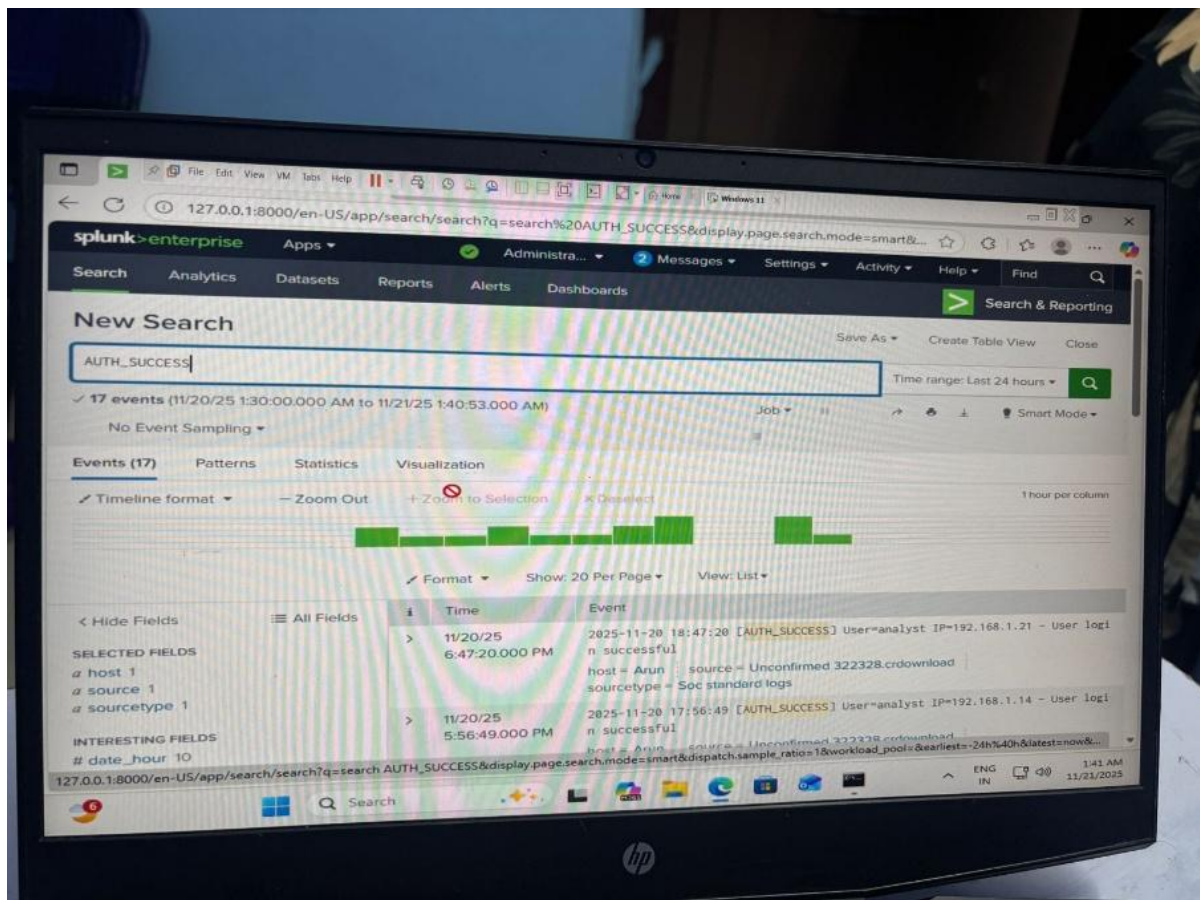
Summary of Logs Ingested

Splunk successfully ingested 100 log events across the following categories:

1. AUTH_SUCCESS – Successful authentication attempts

Alert Analysis & Findings

17 successful authentication attempts were logged, providing visibility into valid user activity. These logs help correlate successful access with preceding failures to determine whether any account was compromised.



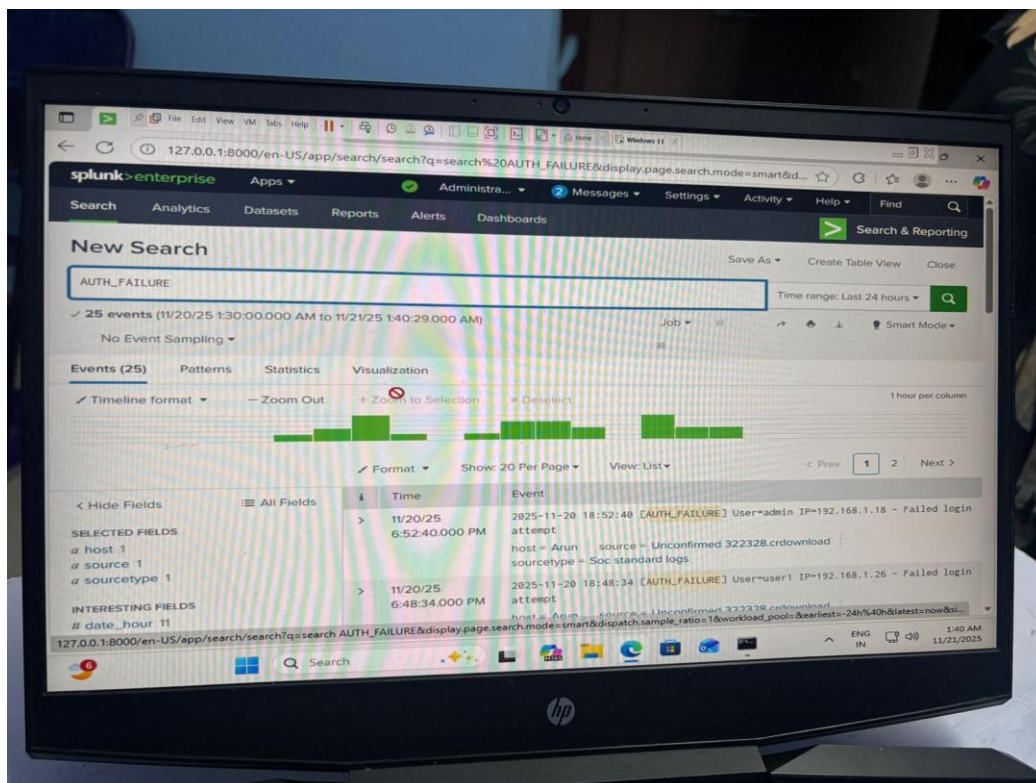


2. AUTH_FAILURE – Failed login attempts (potential brute-force or unauthorized access)

Alert Analysis & Findings

25 failed authentication attempts were observed in the log dataset. Certain IP addresses repeatedly attempted to log in, which may indicate brute-force attempts or unauthorized probing.

- Impact: Risk of account compromise and unauthorized access.
- Evidence: Multiple consecutive AUTH_FAILURE logs from same IP.
- Action Required: Lock affected user accounts, enforce MFA, block suspicious IP addresses.



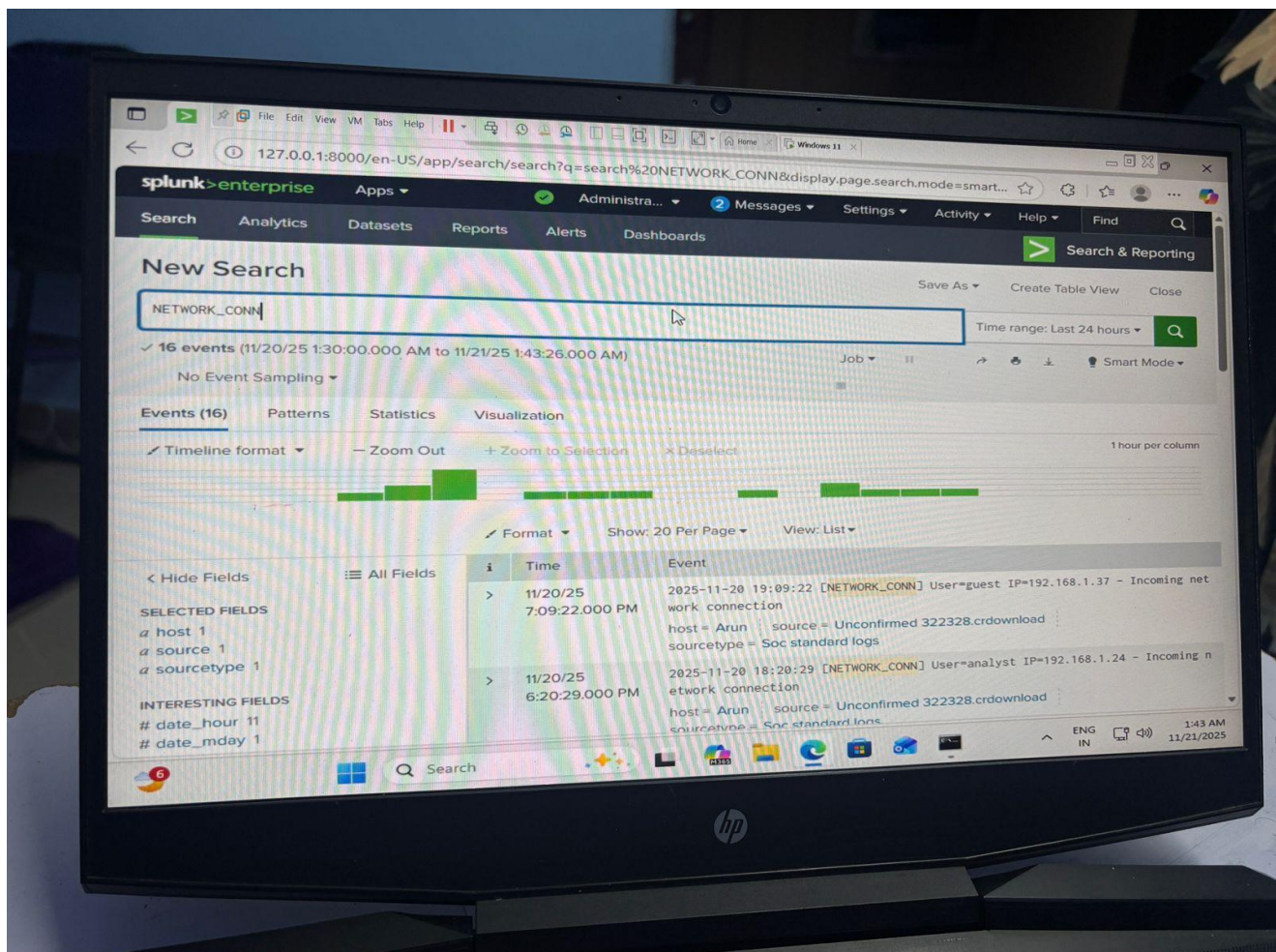


3. NETWORK_CONN – Incoming and outgoing network connections

Alert Analysis & Findings

16 'NETWORK_CONN' events were detected showing inbound connections from internal IP addresses. These logs help trace lateral movement attempts or unusual connections.

- Impact: Potential unauthorized access to internal assets.
- Action Required: Validate user activity, analyze destination ports, check for anomalous traffic volume.





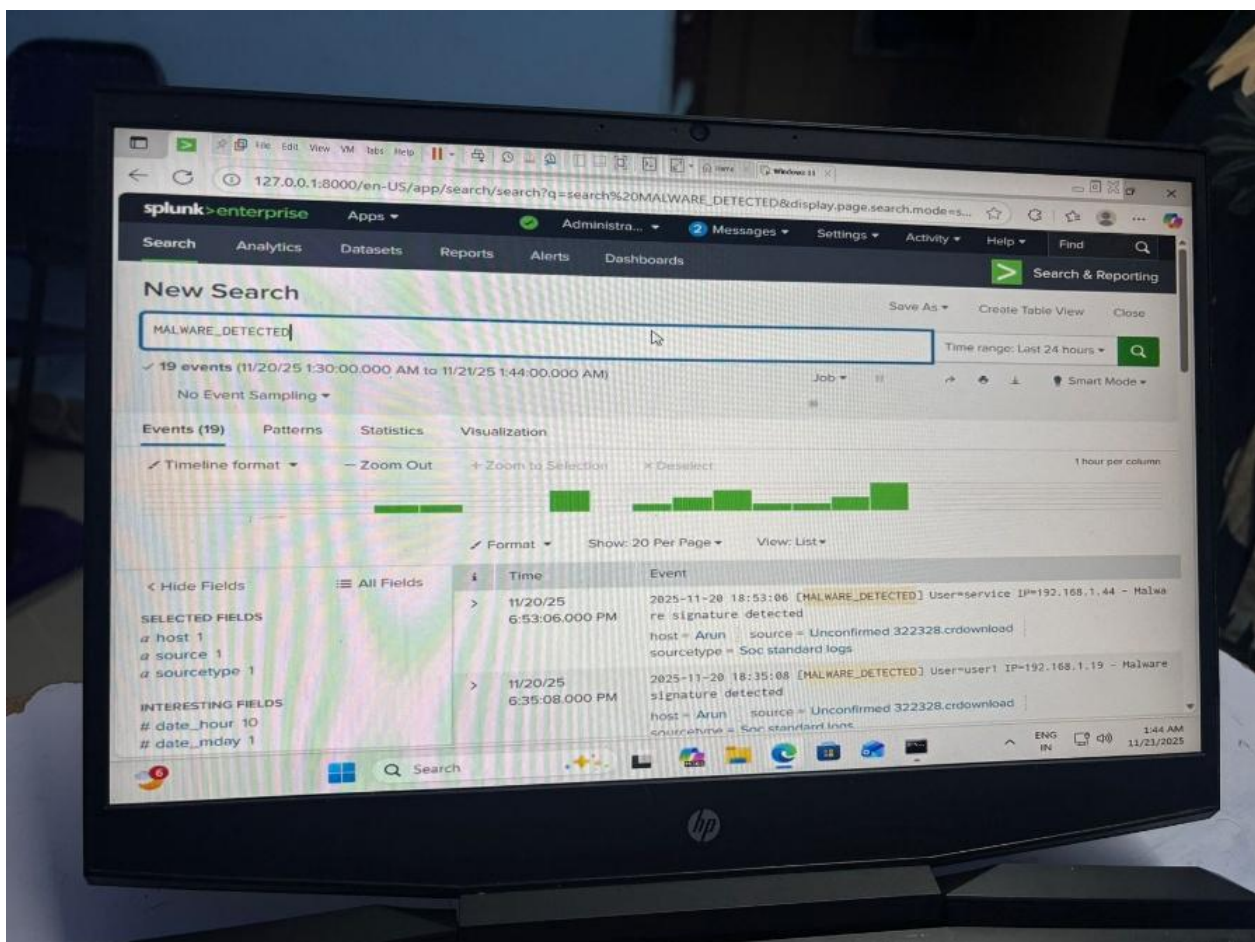
4. MALWARE_DETECTED – Malware signatures detected on endpoints

These logs were parsed, indexed, and analyzed using SPL (Search Processing Language).

Alert Analysis & Findings

Splunk identified multiple 'MALWARE_DETECTED' events across different hosts. The detection logs indicate that a known malware signature was triggered. These events represent high-severity threats requiring immediate action.

- Impact: Possible compromise of endpoint systems.
- Evidence: Multiple malware alerts triggered by user accounts.
- Action Required: Quarantine affected system, run AV scan, block IOC signatures.





Incident Classification

Based on SOC triage guidelines, the following priority mapping applies:

High Severity:

- Malware detections
- Multiple authentication failures

Medium Severity:

- Unusual network connections

Low Severity:

- Normal user logins (AUTH_SUCCESS)

Timeline of Events

Events were logged between 20-Nov-2025 and 21-Nov-2025. Malware detections and authentication failures peak around the evening hours, corresponding to potential targeted attack attempts.

Recommended Mitigation Measures

- Implement Multi-Factor Authentication (MFA)
- Block malicious IP addresses at firewall
- Deploy Endpoint Detection & Response (EDR) tools
- Enforce strong password policies
- Conduct malware scans on affected systems
- Review network segmentation and endpoint hardening

Conclusion

The SOC analysis confirms multiple suspicious activities including malware detection, authentication failures, and unusual network connections. These indicators strongly suggest attempted intrusion and endpoint compromise. Proper mitigation actions must be taken immediately to secure the environment.



SOC Incident Response Report

Timeline of Security Events

Timestamp	Event Type	IP Address	Description
2025-11-20 18:53:06	MALWARE_DETECTED	192.168.1.44	Malware signature detected for user=service
2025-11-20 19:09:22	NETWORK_CONN	192.168.1.37	Incoming network connection detected for user=guest
2025-11-20 18:47:20	AUTH_SUCCESS	192.168.1.21	User login successful for user=analyst
2025-11-20 18:52:40	AUTH_FAILURE	192.168.1.18	Failed login attempt for user=admin
2025-11-20 18:53:08	MALWARE_DETECTED	192.168.1.19	Malware signature detected for user=user1

